

Evaluation Report — Product Intelligence Pipeline

Measured on the organisers' 200-row labelled dataset. Holdout fold: 66 rows the knowledge base was never fitted on (split is a SHA-256 hash of the part number, identical on every machine). Full catalogue: 200 rows. Sources: `metrics-holdout-20260816-020040.json`, `metrics-all-20260817-013153.json`.

1 · Headline results

Every number below is computed by `scripts/run_pipeline.py` against the reference Delivery Format workbook and written to a metrics JSON; this report only reformats those files. Reproduce with `python scripts/run_pipeline.py --fold holdout`.

Metric	Holdout (66 rows)	Full catalogue (200 rows)
Mean exact match, 10 scored fields	40.4%	47.0%
Mean fuzzy match, 10 scored fields	72.7%	84.9%
House-rule compliance (5 rules)	100%	100%
Delivery headers vs Expected Output	252 / 252 (exact)	252 / 252 (exact)
Columns populated	148 / 252	148 / 252
Schema coverage vs human-filled cells	71.2%	79.4%
Attribute label precision	93.2%	97.6%
Rows routed to human review	7 (10.6%)	7 (3.5%)

The holdout column is the honest measurement (train-fold rows are excluded); the full-catalogue column is in-sample for 134 of 200 rows and is quoted only because the delivered dataset is all 200 rows.

2 · Field-level accuracy

Holdout columns score rows never seen by any registry — the honest skill estimate. **All 200** columns score the delivered dataset, which is in-sample for 134 rows and is quoted for completeness. Exact match requires byte equality after normalisation; fuzzy match (RapidFuzz token-set, ≥ 97 for identity fields) credits answers that differ only by word order.

Field	Exact holdout	Fuzzy holdout	Exact all 200	Fuzzy all 200
Manufacturer part number	100.0%	100.0%	99.0%	99.5%
Manufacturer name	89.4%	92.4%	95.0%	96.0%
Brand name	89.4%	90.9%	95.0%	95.5%
Classpath (taxonomy)	74.2%	75.8%	91.5%	92.0%
Product name	47.0%	72.7%	81.0%	91.0%
Long description (product page)	0.0%	69.7%	0.0%	85.5%
Short description (search)	0.0%	62.1%	1.0%	80.0%
Retail description (label)	0.0%	63.6%	1.0%	85.0%
Mobile description (60–80 chars)	4.5%	50.0%	6.0%	65.0%
Invoice description (≤ 40 chars, caps)	0.0%	50.0%	0.5%	60.0%

3 · House-rule compliance

Measured on every emitted row regardless of ground truth, so it also holds for catalogues with no labels at all.

House rule	Holdout	Full catalogue
Invoice $\leq$ 40 characters	100.0%	100.0%
Invoice ALL CAPS	100.0%	100.0%
Mobile within 60–80 characters	100.0%	100.0%
Unit spacing (24 in, never 24in)	100.0%	100.0%
Fractions not decimals (50-1/4 in)	100.0%	100.0%

4 · Attribute grid

Labels are constrained to the controlled vocabulary reconstructed from the labelled rows (317 labels, 1,385 values); values are snapped onto observed spellings. Precision counts emitted labels that exist in the category template.

Measure	Holdout	Full catalogue
Attribute labels emitted	1,397	4,028
Label precision	93.2%	97.6%
Attribute values compared to reference	286	1,263
Value accuracy (labelled rows)	59.8%	70.0%

5 · Traceability — first-party retrieval

The sourcing rule is enforced by a fail-closed domain policy: only the resolved manufacturer's own site (or a known official document CDN) is read; marketplaces and distributor sites are refused before the request and again after redirects. A value taken from the web is only cited when it appears **verbatim** in the fetched document.

Measure	Holdout	Full catalogue
Rows with a verified first-party source	14 / 66	50 / 200
Deep product links in MFR URL	14 / 66	50 / 200
Documents read (pages + manuals)	22	80
Attribute values grounded verbatim to a URL	88 (26.1% of filled)	348 (23.9% of filled)
Requests blocked by policy / robots.txt	0 / 0	—

Reach is limited by bot protection on manufacturer sites (Satco returns 429, GE Appliances and others 403); those walls are respected, not worked around. Where no first-party page is reachable the row degrades to input-only behaviour rather than borrowing reseller data.

6 · Throughput, cost and resilience

Measure	Scored run (live web)	Delivered run (cache replay)
Rows	66	200
Wall clock	1.4 s	10.4 s
Per row	0.02 s	0.05 s
Model calls (live / cached)	0 / 161	0 / 432
Live tokens	0	0
Model failures	0	0
Stage failures	0	0

The scored run included live retrieval (122 requests, 68 MB fetched). Worst-case cold-cache bill measured on the same fold: 47,360 tokens $\approx$ a fraction of a cent per row at small-model pricing; the prototype ran on free tiers.

7 · Limitations, stated plainly

- **Description exact-match is near zero on the holdout.** Reference descriptions contain specifications (amperage, drum material, annual energy) that appear nowhere in the 15-character input; they exist only in manufacturer documentation. The pipeline reproduces structure, ordering and house style (84–93% similarity) and refuses to invent the content.
- **First-party retrieval reaches 21% of holdout rows.** The ceiling is bot protection, not discovery. Unsourced rows keep the verified manufacturer domain rather than a fabricated deep link.
- **EAN, UNSPSC and TRADE_NAME are left blank.** They cannot be derived from the input; UPC/GTIN are written only when the check digit validates.
- **Registries are fitted on 134 rows.** 62 classpaths and 38 brands is thin; several surfaces are learned from one example. More labelled data improves this with no code change.
- **Full-catalogue accuracy is in-sample for 134 of 200 rows** and is quoted for the delivered artefact, not as an honest skill estimate — the holdout column is that estimate.

Reproduce everything: `python scripts/run_pipeline.py --fold holdout` (scores + metrics JSON), `python scripts/verify_schema.py` (252-header compliance gate, exits non-zero on drift), `python scripts/build_report.py` (this document). Tests: 182, no network, no API key (`python -m pytest tests -q`).